

CHAPTER 3 – PART 3

SOLVING PROBLEMS BY SEARCHING

INTRODUCTION TO ARTIFICIAL INTELLIGENT (AI)



REV: 06.04.2023

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Informed (Heuristic) Search

Unlike uninformed search algorithms, **informed (heuristic) search** algorithms consider problem-specific knowledge; thus, they can find solutions more efficiently.

Informed search algorithms use the **best-first search** approach. That is; in the frontier, the nodes to be expanded are chosen according to **an evaluation function $f(n)$** . The implementation of best-first search is very similar to that for uniform-cost search, except it uses the **evaluation function f** instead of **path cost g** to order the priority queue.

$f(n) = ?$



The choice of **f** determines the search strategy.

n : node

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Heuristic Function

Evaluation functions $f(n)$ include a heuristic function $h(n)$.

$g(n)$



The cost of the path from the initial node to the state at node n .

$h(n)$



An estimated cost of the cheapest path from the state at node n to a goal state

We will deal with heuristic functions, which are

- non-negative,
- problem-specific.
- $h(n) = 0$ if n is a goal node

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Here, two best-first search strategies are presented:

1. **Greedy best-first search**
2. **A* (A-star) search**

**Greedy best-first
search**



$$f(n) = h(n)$$

A* search



$$f(n) = g(n) + h(n)$$

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Greedy Best-First Search

Greedy (best-first) search tries to expand the node that is closest to the goal to reach a solution quickly. Thus, it evaluates nodes by using just the heuristic function; that is

$$f(n)=h(n)$$

h_{SLD}



The Straight Line Distance Heuristic is one of the most used heuristic functions.

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The **tree-search** algorithm for **greedy best-first search** is given below.

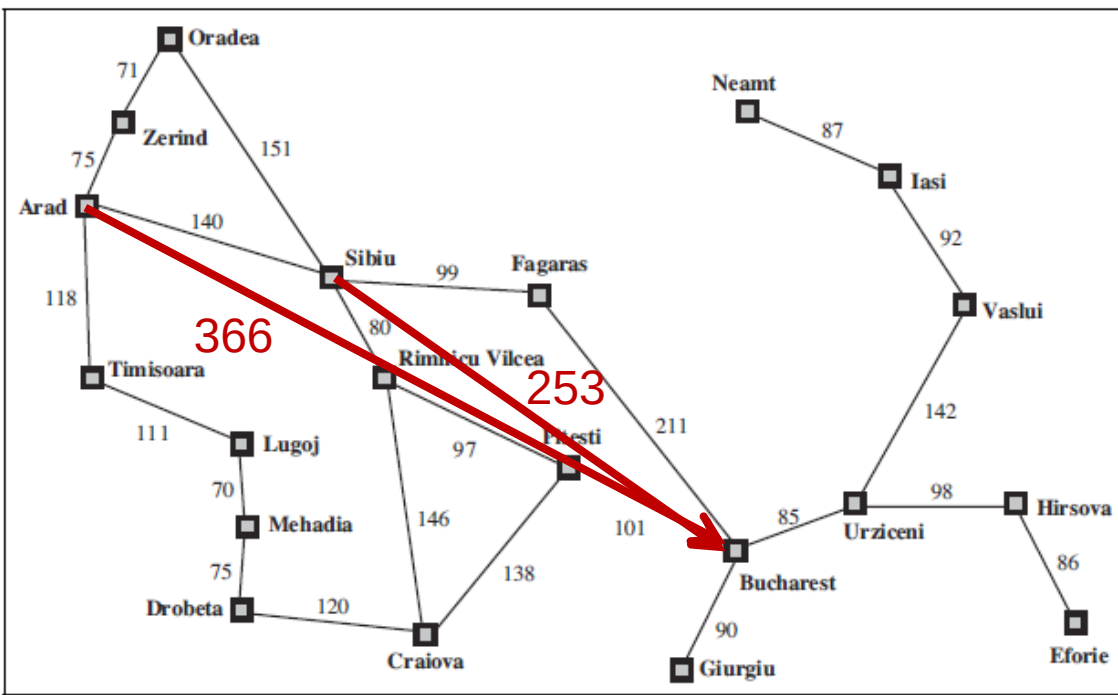
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function GREEDY-BEST-FIRST-SEARCH-TREE (problem) returns a solution, or failure
- node ← a node with STATE = problem.INITIAL-STATE (including heuristic function),
  PATH-COST = 0
- if problem.GOAL-TEST(node.STATE) then return SOLUTION(node)
- frontier ← a priority queue ordered by  $h(n)$ , with the root node
- loop do
  - if EMPTY?( frontier) then return failure
  - node ← POP( frontier ) /* chooses node with min heuristic function in frontier */
  - for each action in problem.ACTIONS(node.STATE) do
    - child ← CHILD-NODE(problem, node, action)
    - if problem.GOAL-TEST(child .STATE) then return SOLUTION(child )
    - frontier ← INSERT(child , frontier )
```

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The Straight-Line Distance Heuristic



A simplified road map of part of Romania.

Arad	366	Mehadia	241
Bucharest	0	Neamt	234
Craiova	160	Oradea	380
Drobeta	242	Pitesti	100
Eforie	161	Rimnicu Vilcea	193
Fagaras	176	Sibiu	253
Giurgiu	77	Timisoara	329
Hirsova	151	Urziceni	80
Iasi	226	Vaslui	199
Lugoj	244	Zerind	374

Values of h_{SLD} —straight-line distances to Bucharest.

The Straight Line
Distance Heuristic

h_{SLD}

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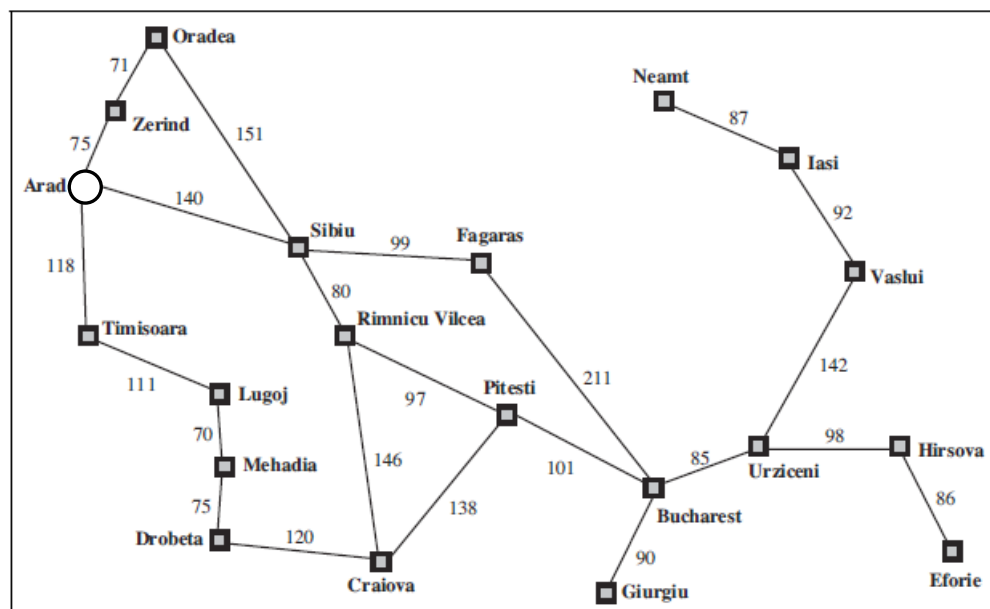
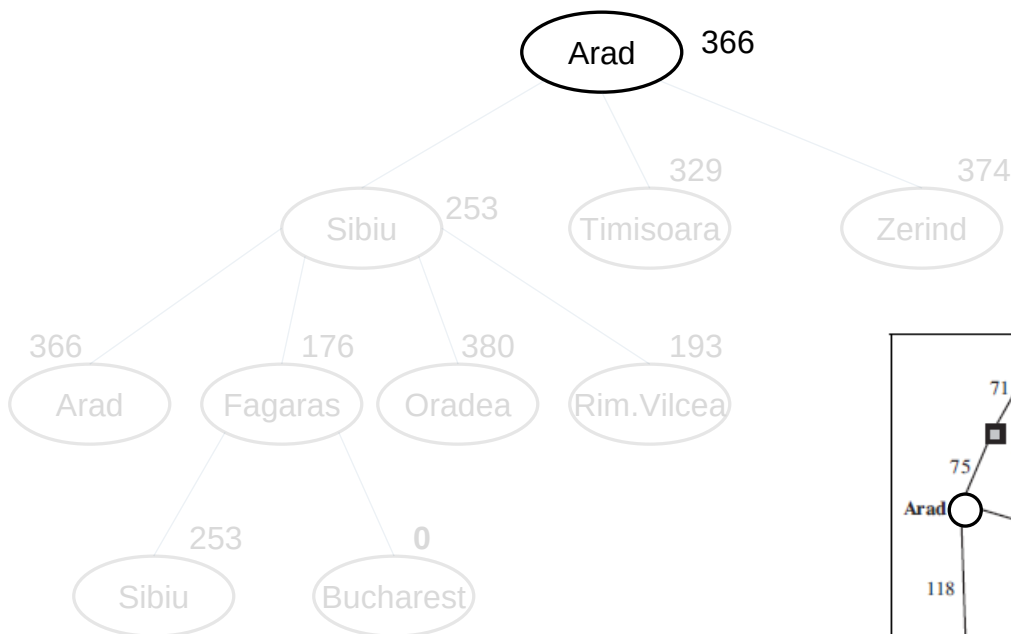
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Ex : The Romania Map

[Arad₃₆₆]

QUEUE



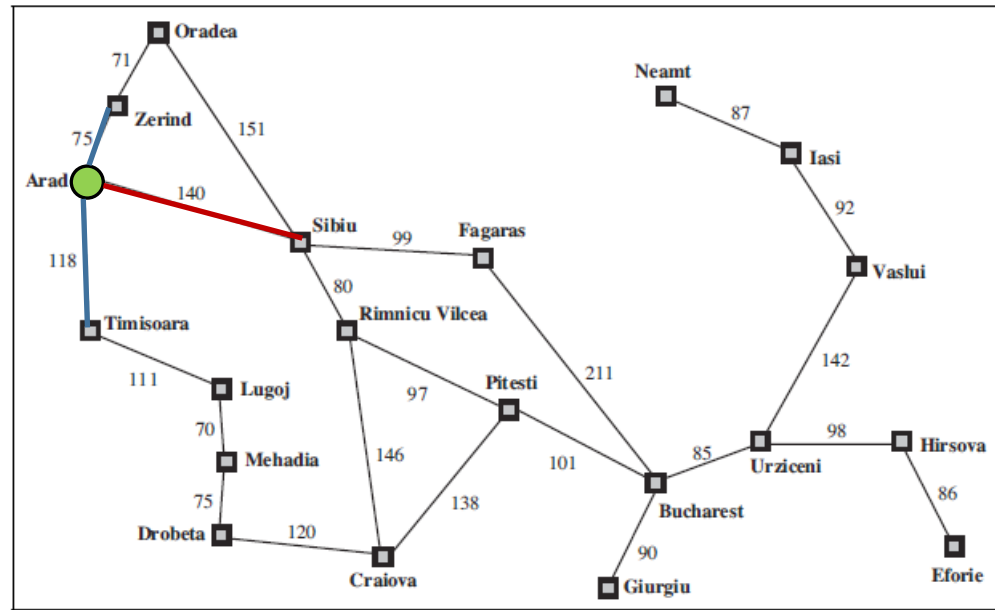
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QUEUE

[Arad₃₆₆]

[**Sibiu**₂₅₃, Timisoara₃₂₉, Zerind₃₇₄]

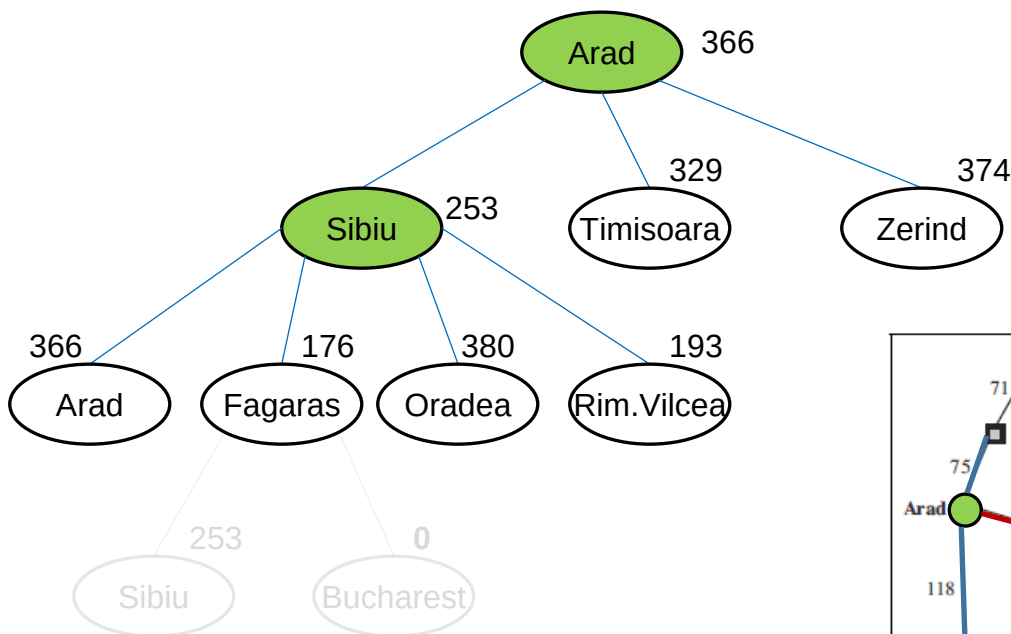


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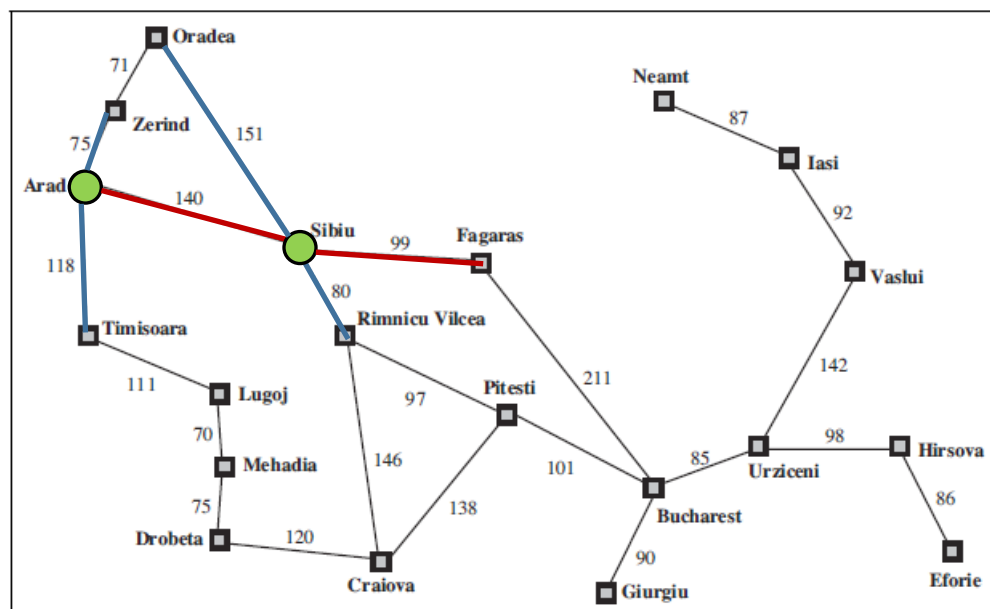


Ex : The Romania Map



QUEUE

[Arad₃₆₆]
[Sibiu₂₅₃, Timisoara₃₂₉, Zerind₃₇₄]
[Fagaras₁₇₆, Rim.Vilcea₁₉₃, Timisoara₃₂₉, Arad₃₆₆,
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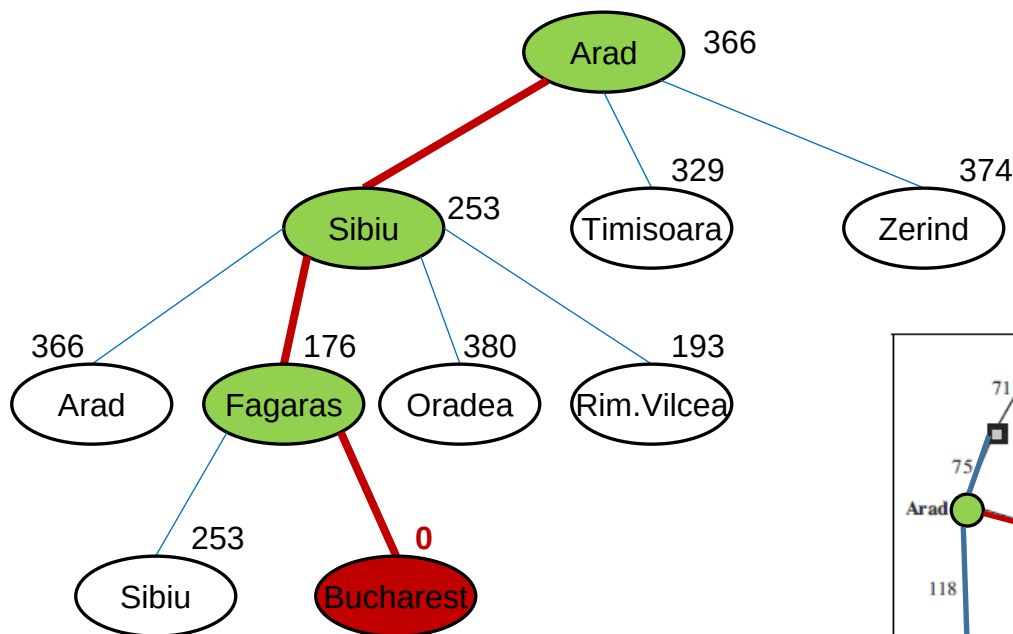


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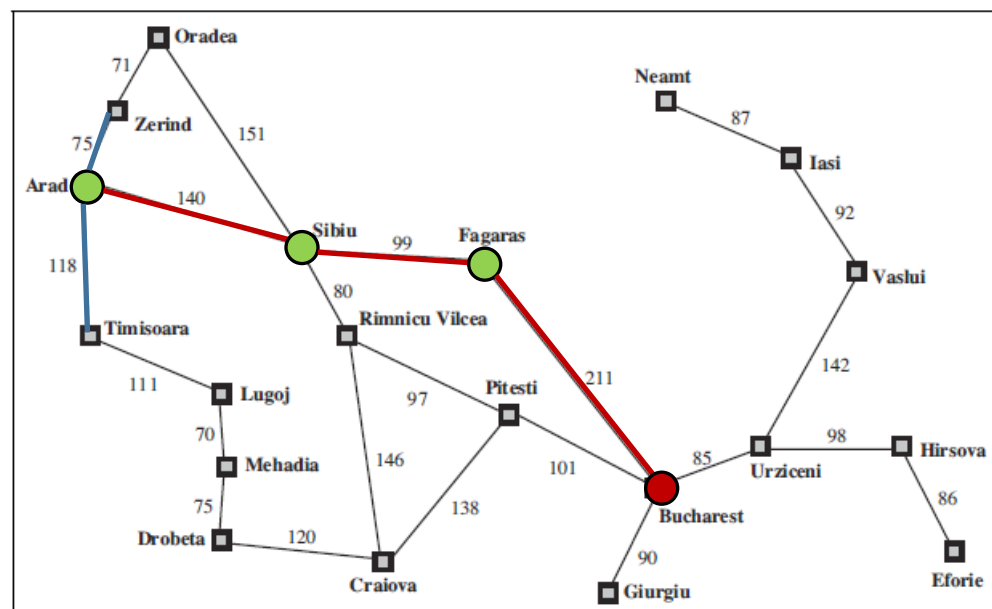


Ex : The Romania Map



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[Rim.Vilcea₁₉₃, Sibiu₂₅₃, Timisoara₃₂₉, Arad₃₆₆,
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The **graph-search** algorithm for **greedy best-first search** is given below.

```
function GREEDY-BEST-FIRST-SEARCH-GRAPH (problem) returns a solution, or failure
- node ← a node with STATE = problem.INITIAL-STATE (including heuristic function),
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- if problem.GOAL-TEST(node.STATE) then return SOLUTION(node)
- frontier ← a priority queue ordered by  $h(n)$ , with the root node
- explored ← an empty set
- loop do
  - if EMPTY?( frontier) then return failure
  - node ← POP( frontier ) /* chooses node with min heuristic function in frontier */
  - add node.STATE to explored
  - for each action in problem.ACTIONS(node.STATE) do
    - child ← CHILD-NODE(problem, node, action)
    - if child .STATE is not in explored or frontier then
      - if problem.GOAL-TEST(child .STATE) then return SOLUTION(child )
      - frontier ← INSERT(child , frontier )
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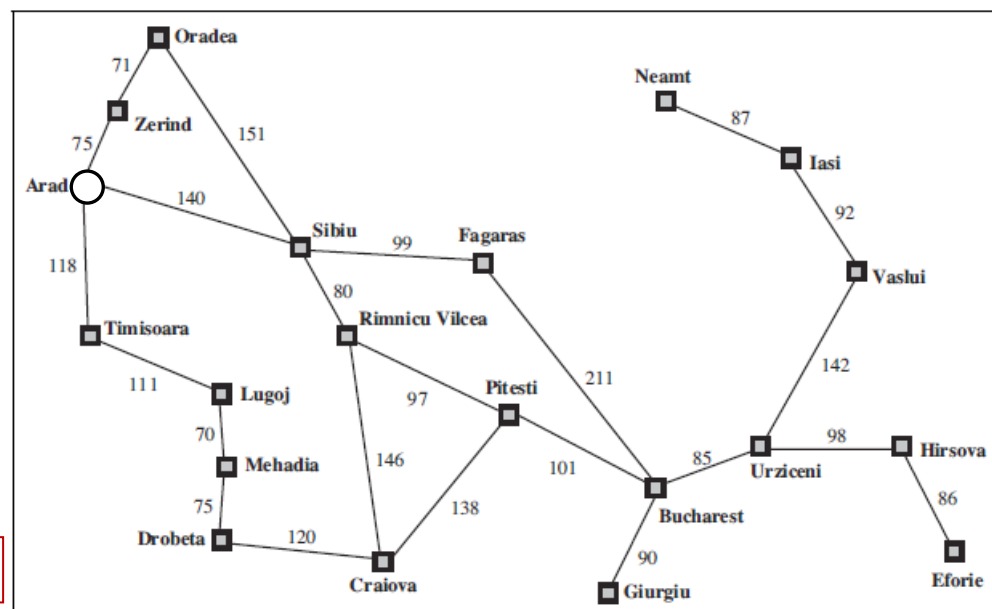
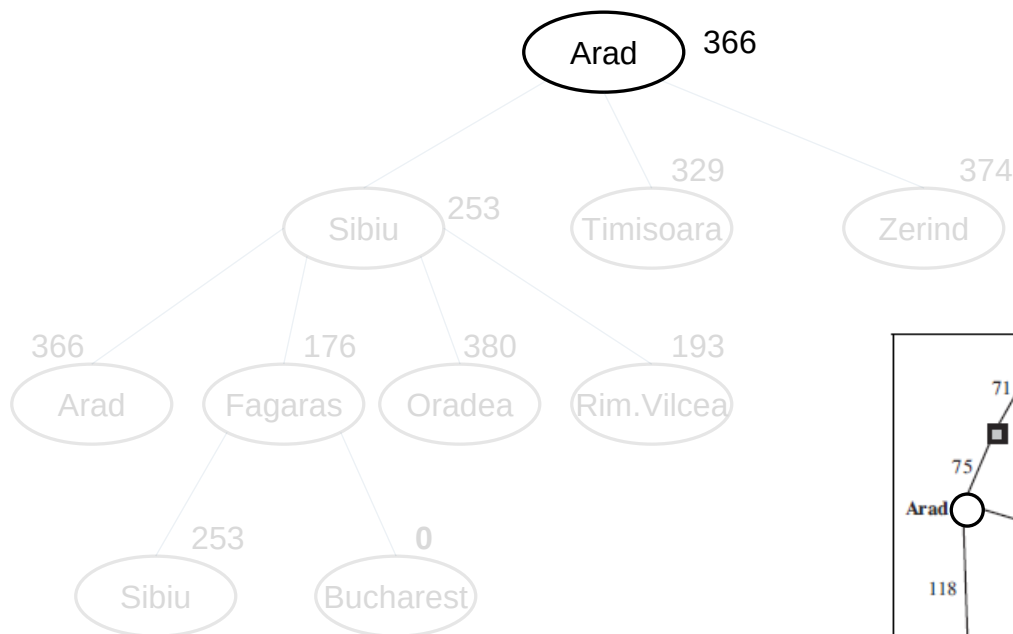
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Ex : The Romania Map

[Arad₃₆₆]

QUEUE



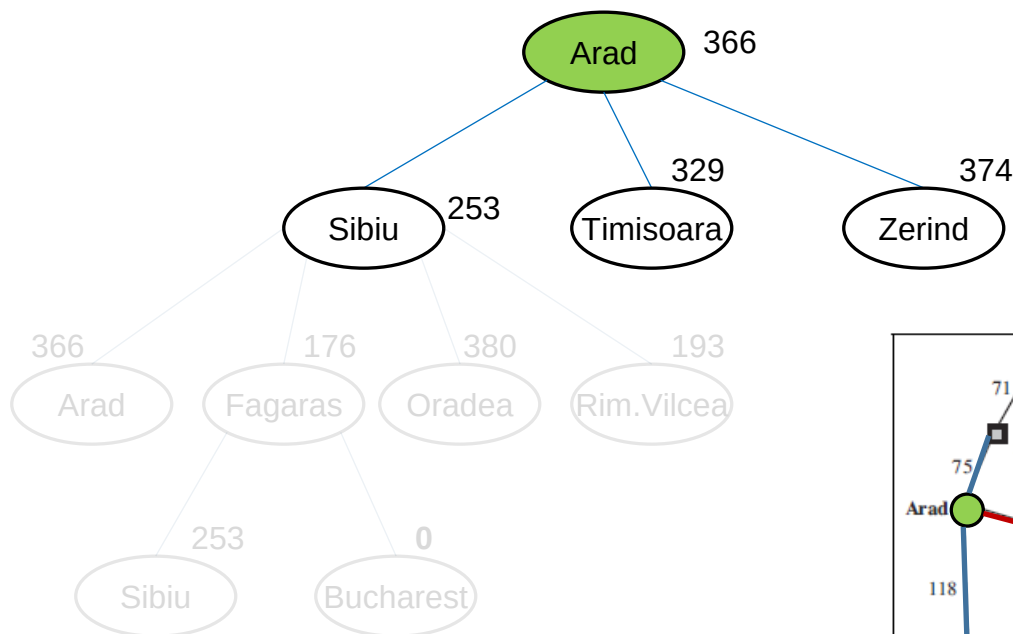
EXPLORED SET:

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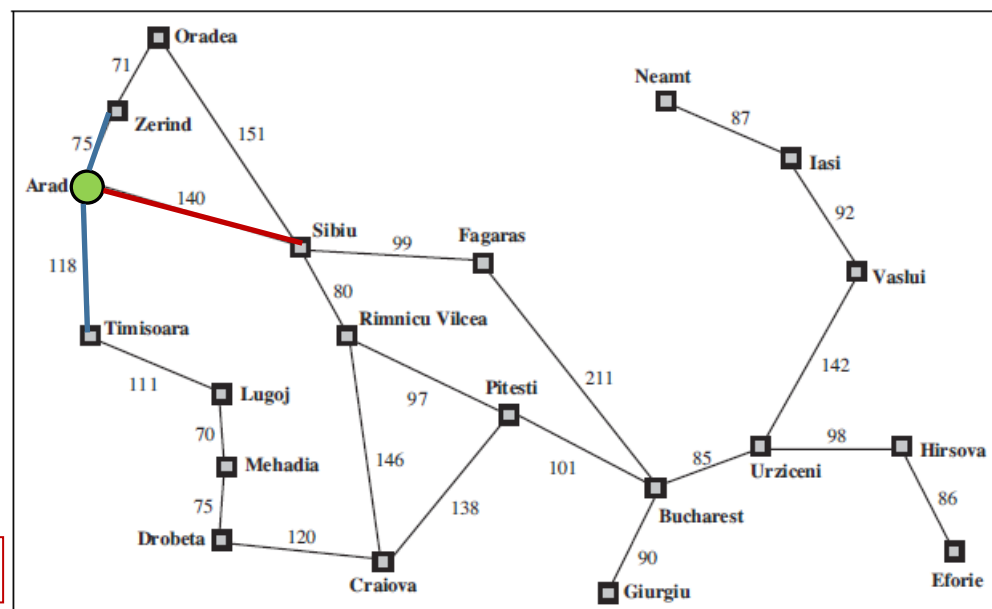
Ex : The Romania Map



[Arad₃₆₆]
[Sibiu₂₅₃, Timisoara₃₂₉, Zerind₃₇₄]

QUEUE

EXPLORED SET: Arad

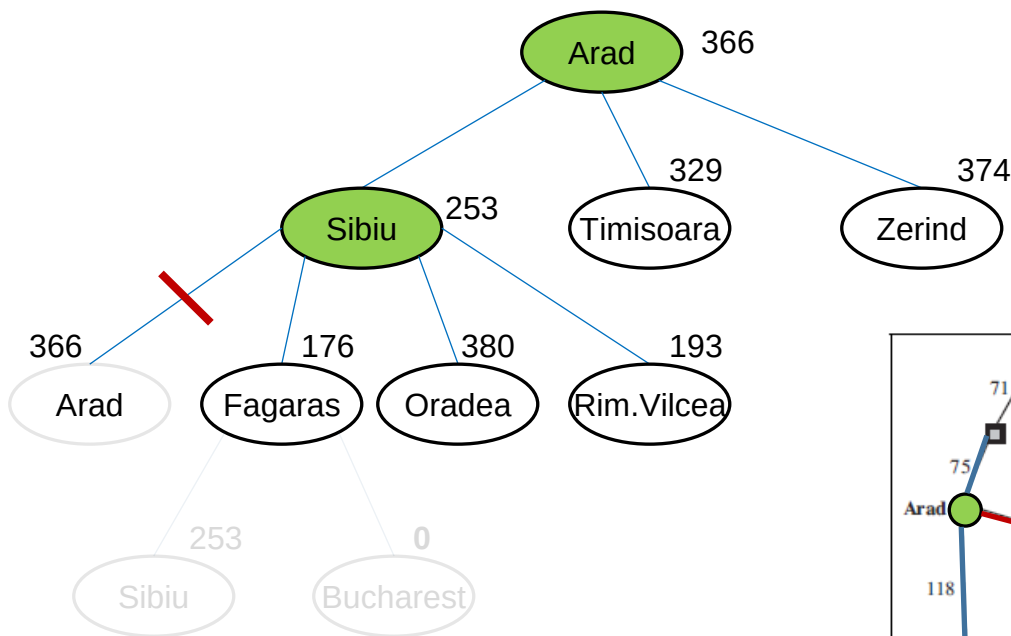


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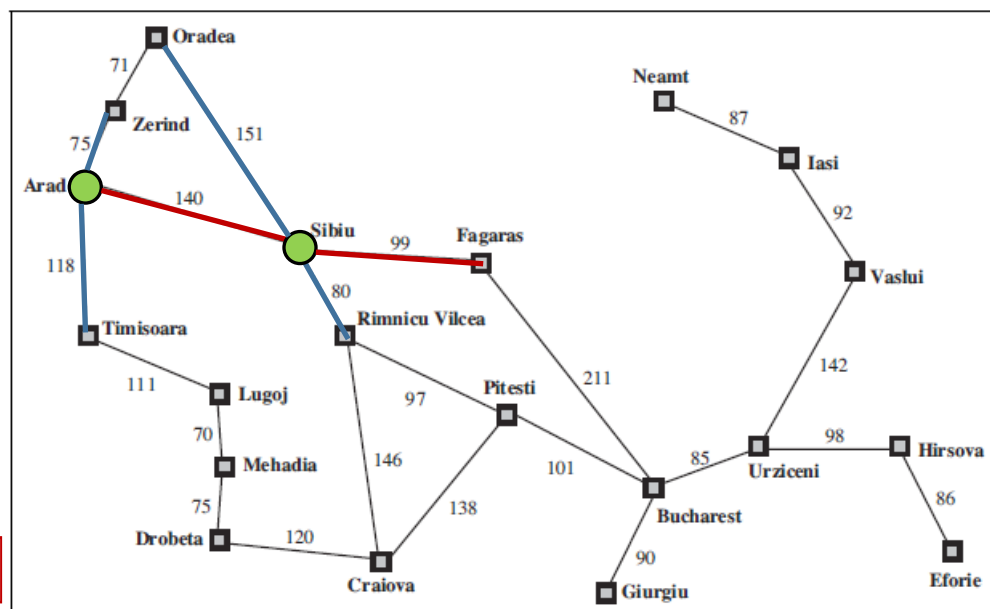
Ex : The Romania Map



[Arad₃₆₆]
[Sibiu₂₅₃, Timisoara₃₂₉, Zerind₃₇₄]
[Fagaras₁₇₆, Rim.Vilcea₁₉₃, Timisoara₃₂₉,
Zerind₃₇₄, Oradea₃₈₀]

QUEUE

EXPLORED SET: Arad, Sibiu

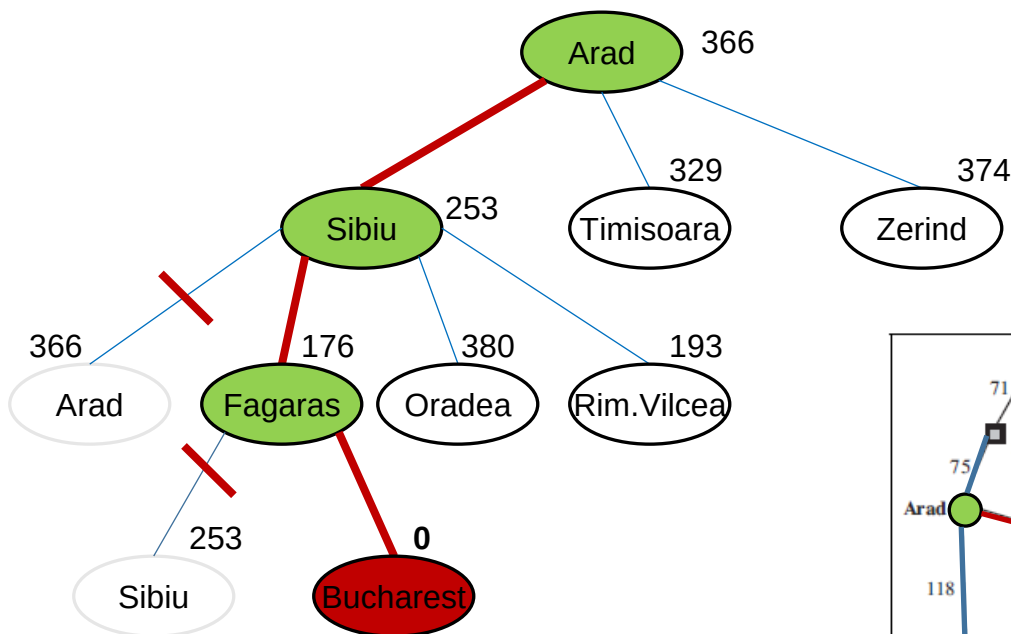


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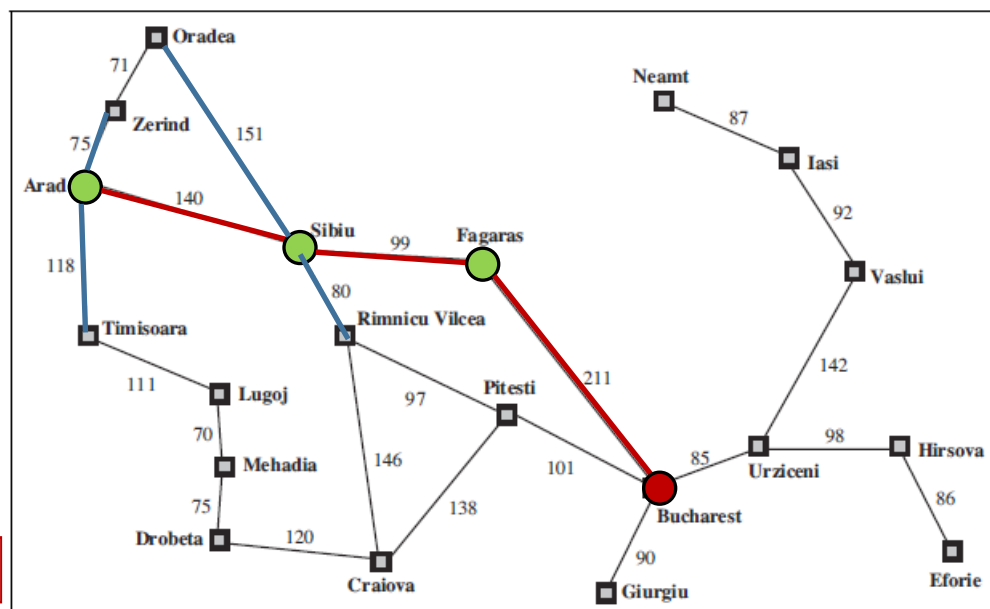
Ex : The Romania Map



QUEUE

[Arad₃₆₆]
[Sibiu₂₅₃, Timisoara₃₂₉, Zerind₃₇₄]
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[Rim.Vilcea₁₉₃, Timisoara₃₂₉,
Zerind₃₇₄, Oradea₃₈₀]

EXPLORED SET: Arad, Sibiu, Fagaras

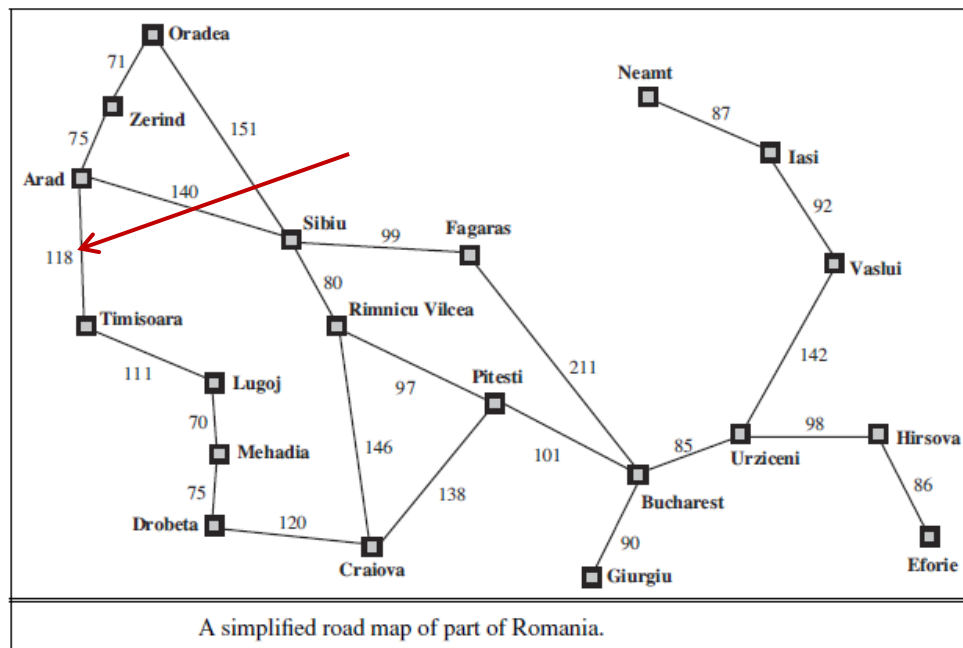


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Greedy Best-First search algorithm's performance



- Greedy best-first tree search is **incomplete** (Consider the solution from Iasi to Fagaras) for tree-search, **complete** (when the state space is finite) for graph-search.
- Greedy best-first search is **not optimal**.

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Greedy Best-First search algorithm's performance?

The **worst-case time complexity and space complexity** is $O(b^m)$, where m is the maximum depth of the search space.

However, with a good heuristic function, the number of searched nodes can be reduced substantially.

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A* Search

A* (A star) search evaluates nodes by combining the path with the heuristic function:

$$f(n) = g(n) + h(n)$$

h_{SLD}



The Straight Line Distance Heuristic is one of the most used heuristic functions.

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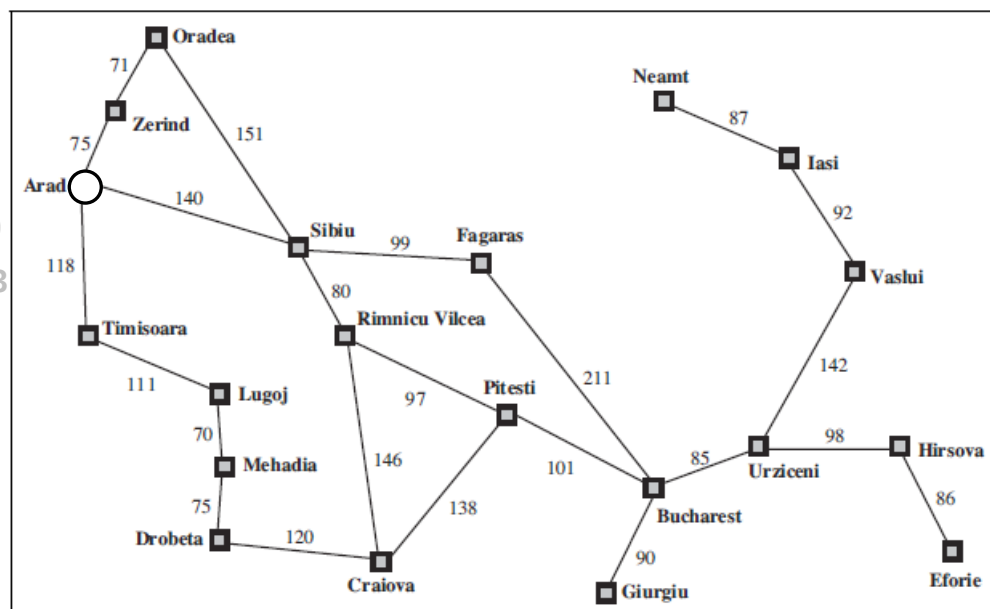
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Ex : The Romania Map

[Arad₃₆₆]

QUEUE



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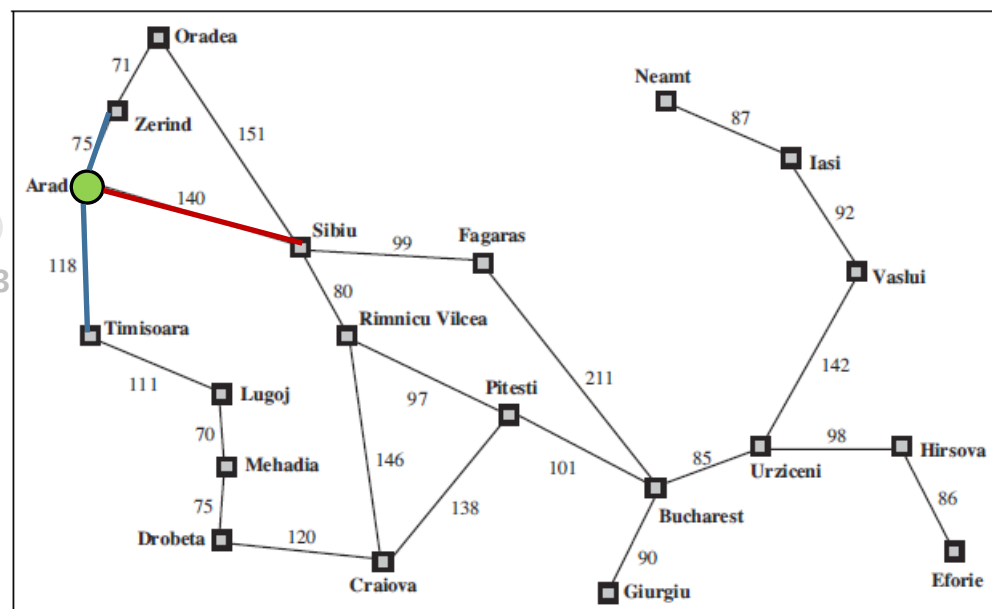
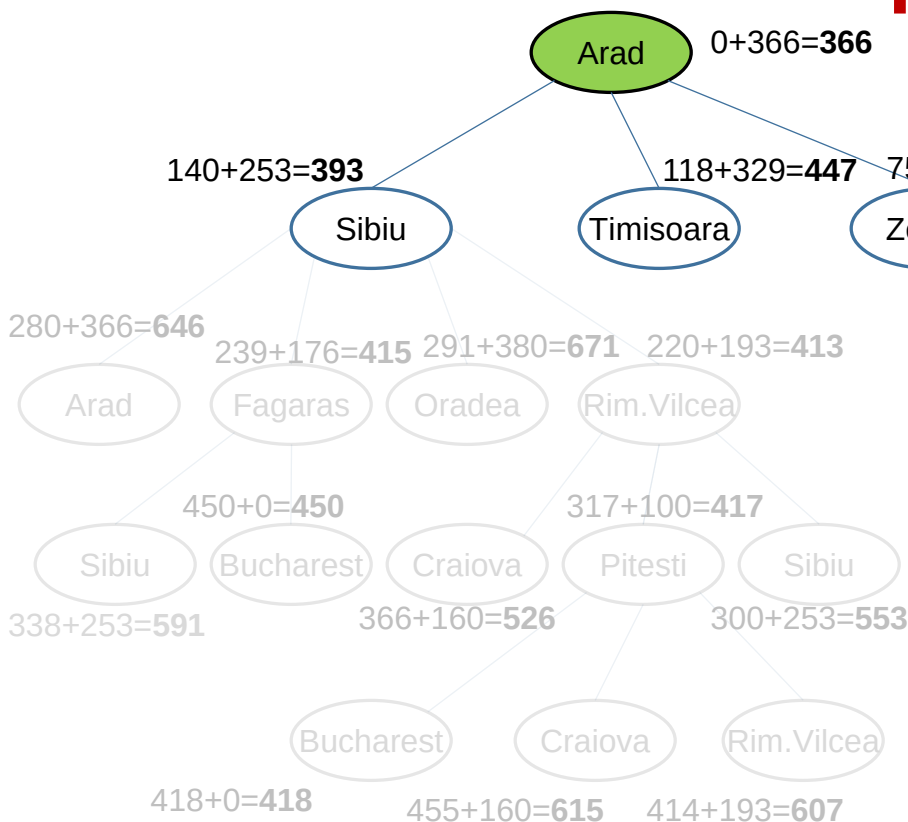
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Ex : The Romania Map

QUEUE

[Arad₃₆₆]
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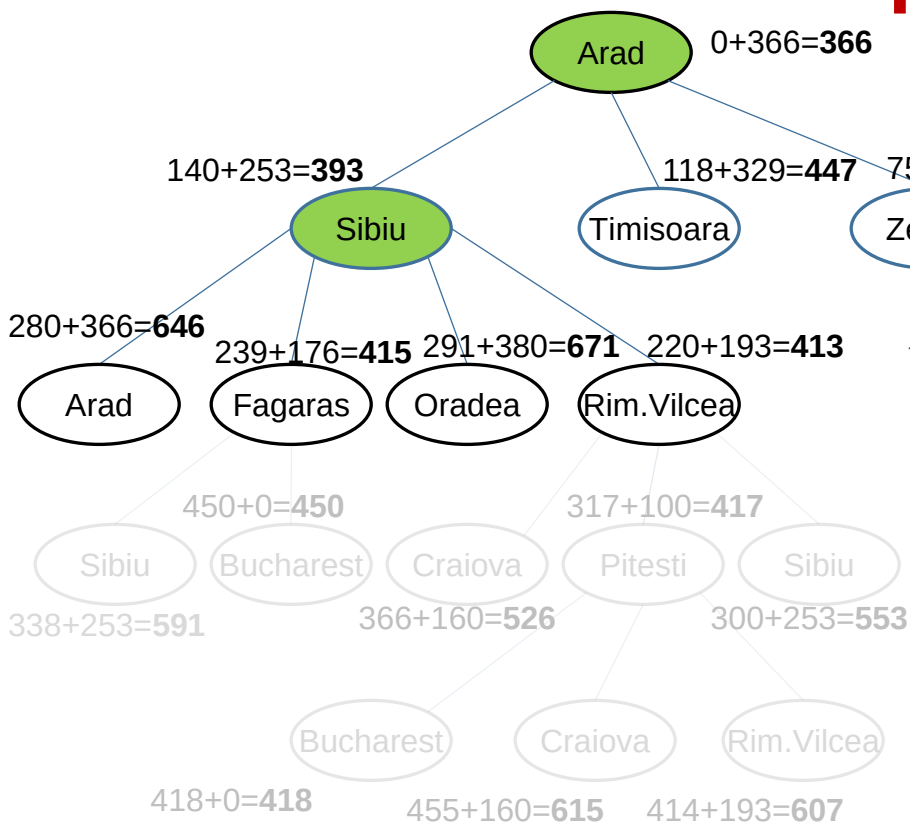


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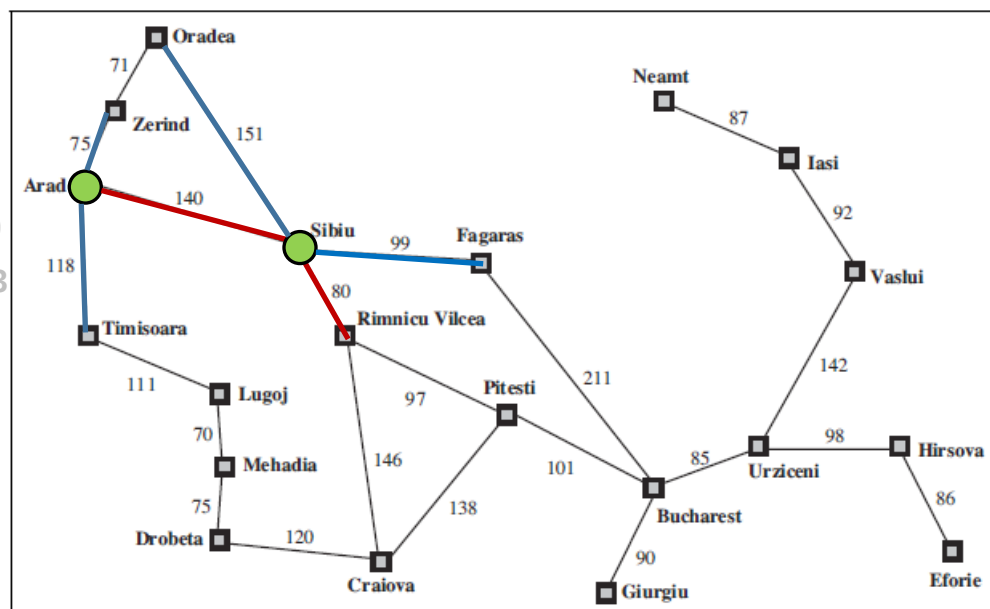


Ex : The Romania Map



QUEUE

[Arad₃₆₆]
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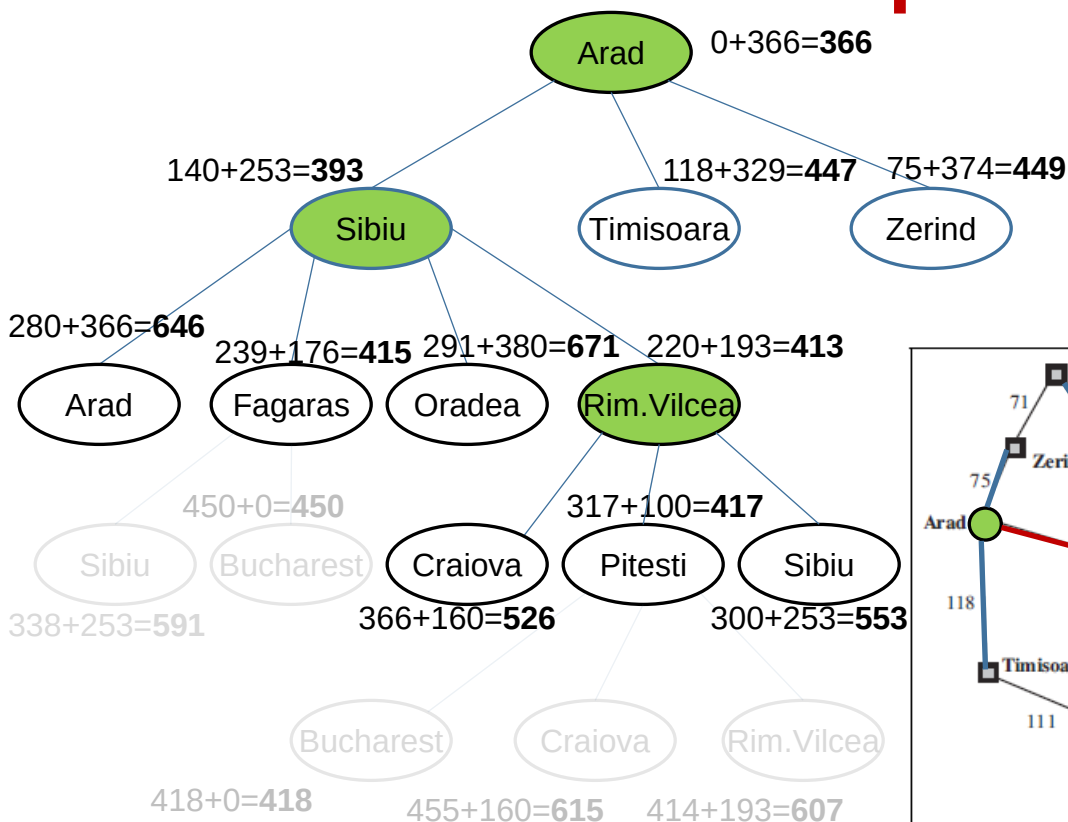


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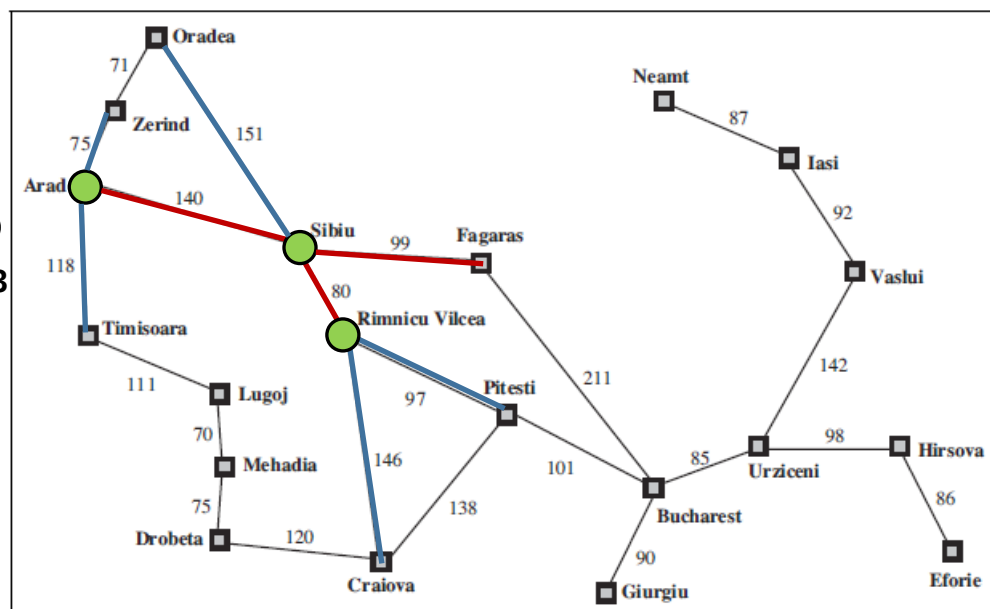


Ex : The Romania Map



QUEUE

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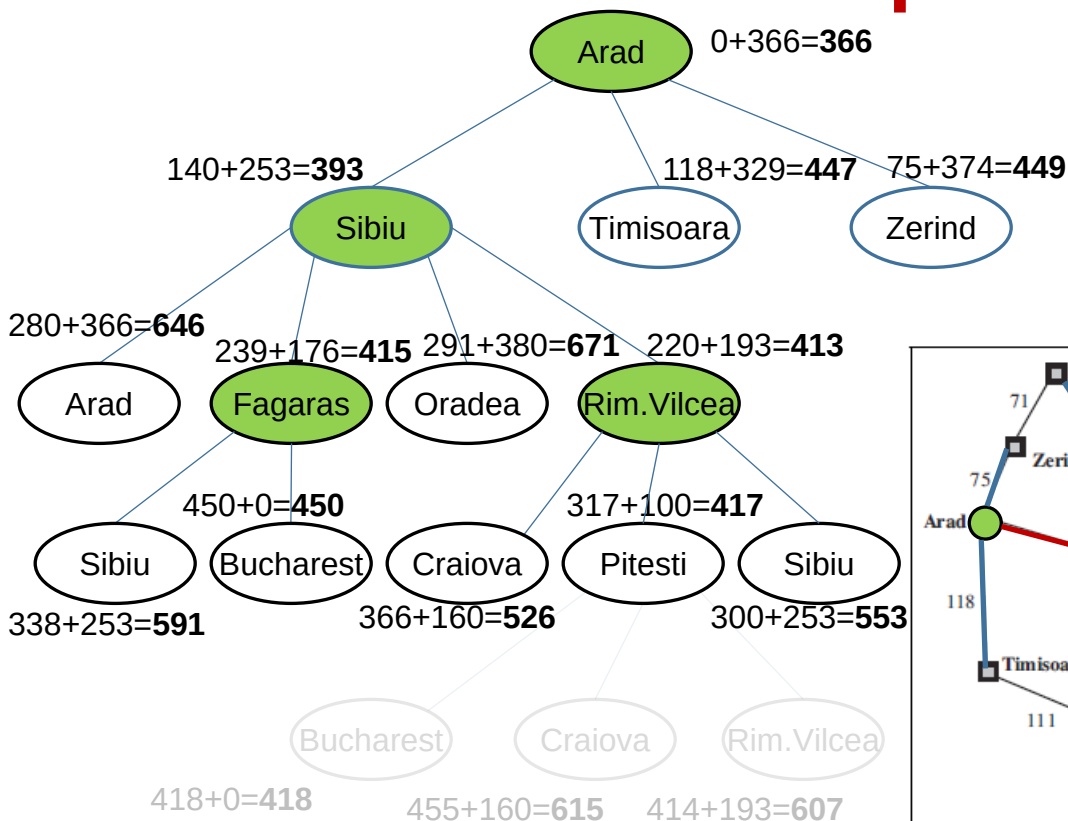


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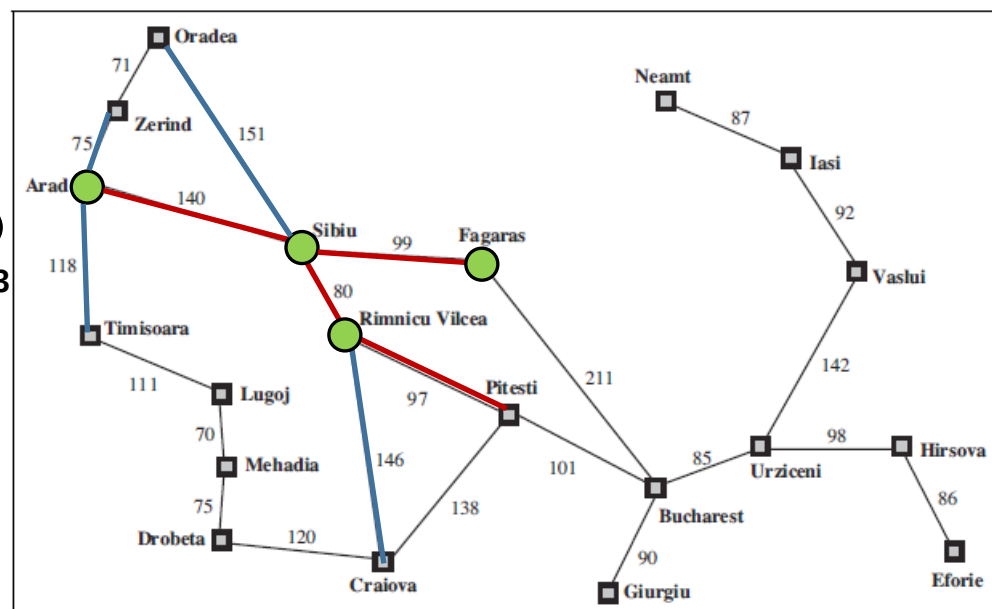


Ex : The Romania Map



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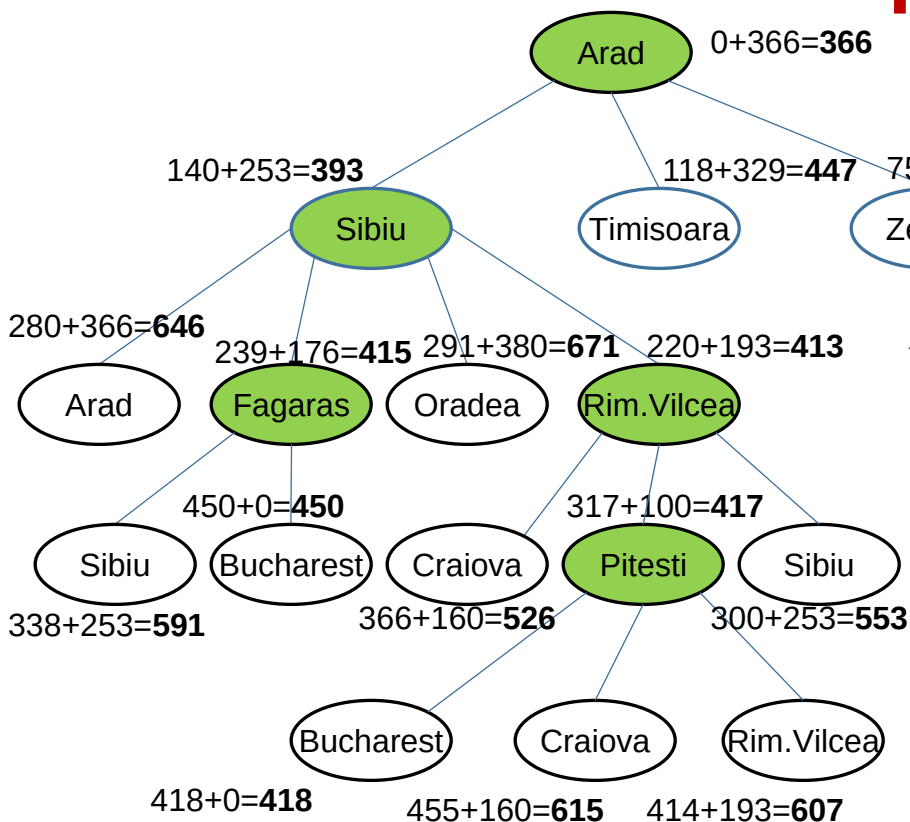


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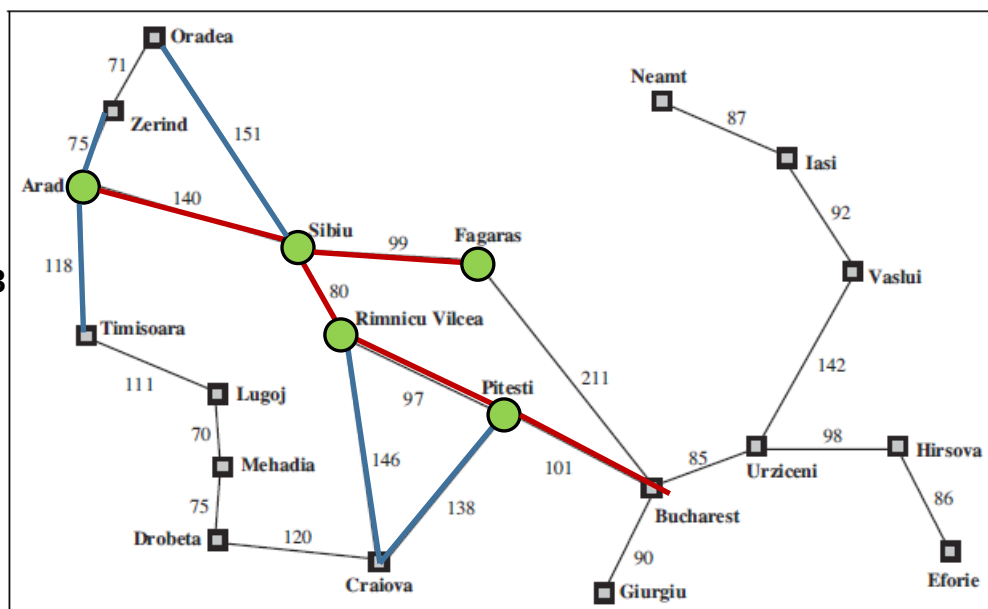


Ex : The Romania Map



QUEUE

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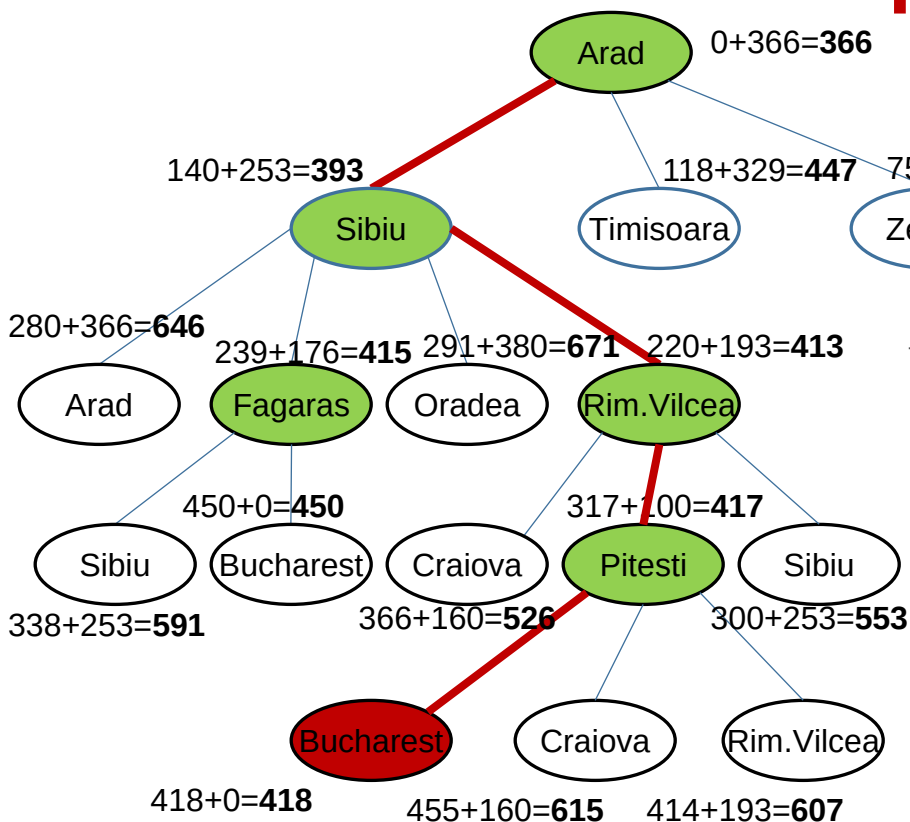


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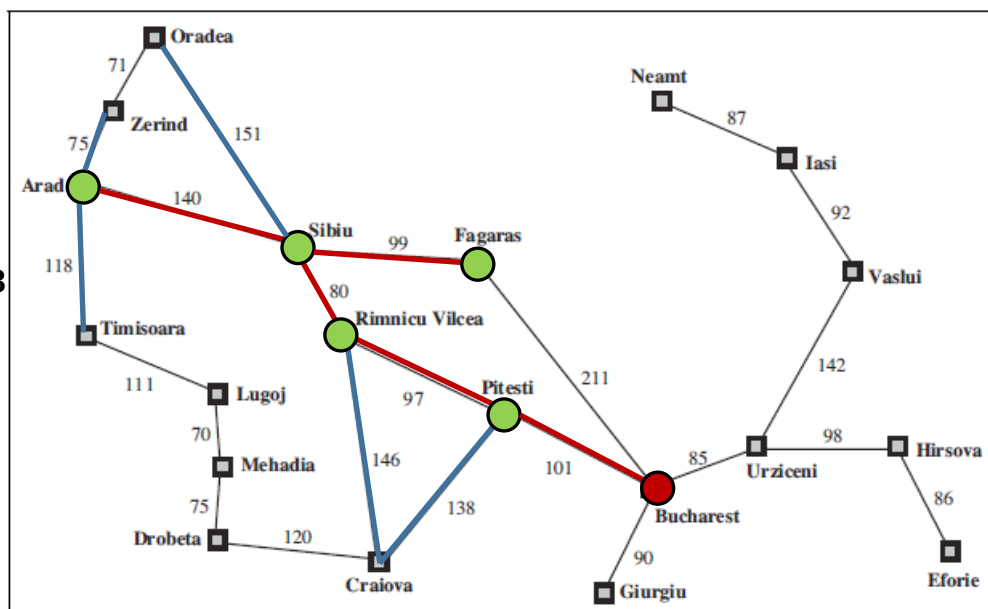


Ex : The Romania Map



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[Arad₃₆₆]
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  - for each action in problem.ACTIONS(node.STATE) do
    - child ← CHILD-NODE(problem, node, action)
    - if child .STATE is not in explored or frontier then
      - frontier ← INSERT(child , frontier )
    else if child .STATE is in frontier with higher  $f(n)$  then
      - replace that frontier node with the child
```

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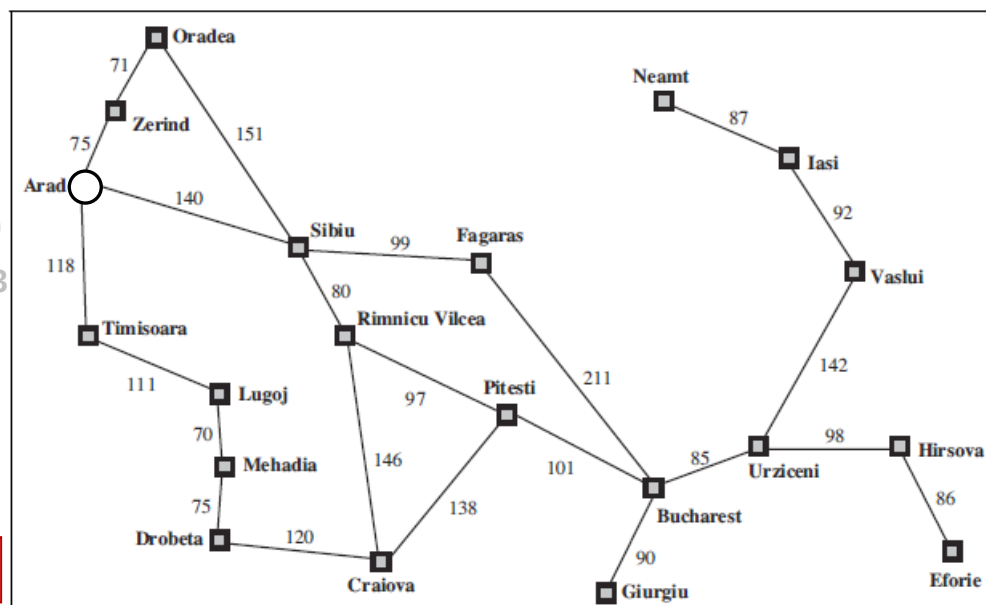
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Ex : The Romania Map

[Arad₃₆₆]

QUEUE



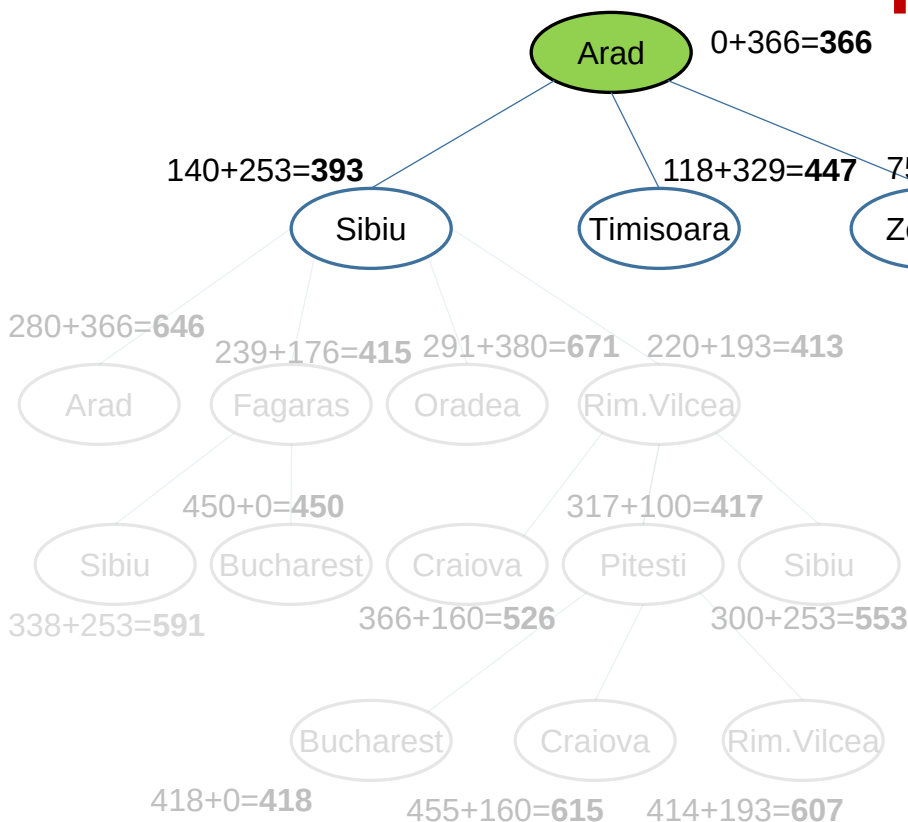
EXPLORED SET:

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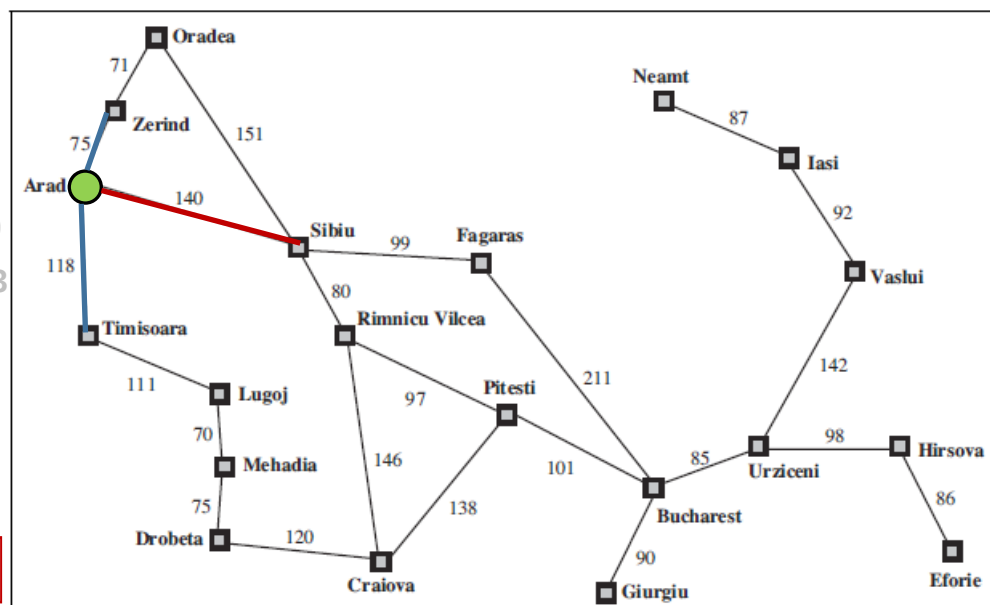
Ex : The Romania Map



EXPLORED SET: Arad

[Arad₃₆₆]
[Sibiu₃₉₃, Timisoara₄₄₇, Zerind₄₄₉]

QUEUE

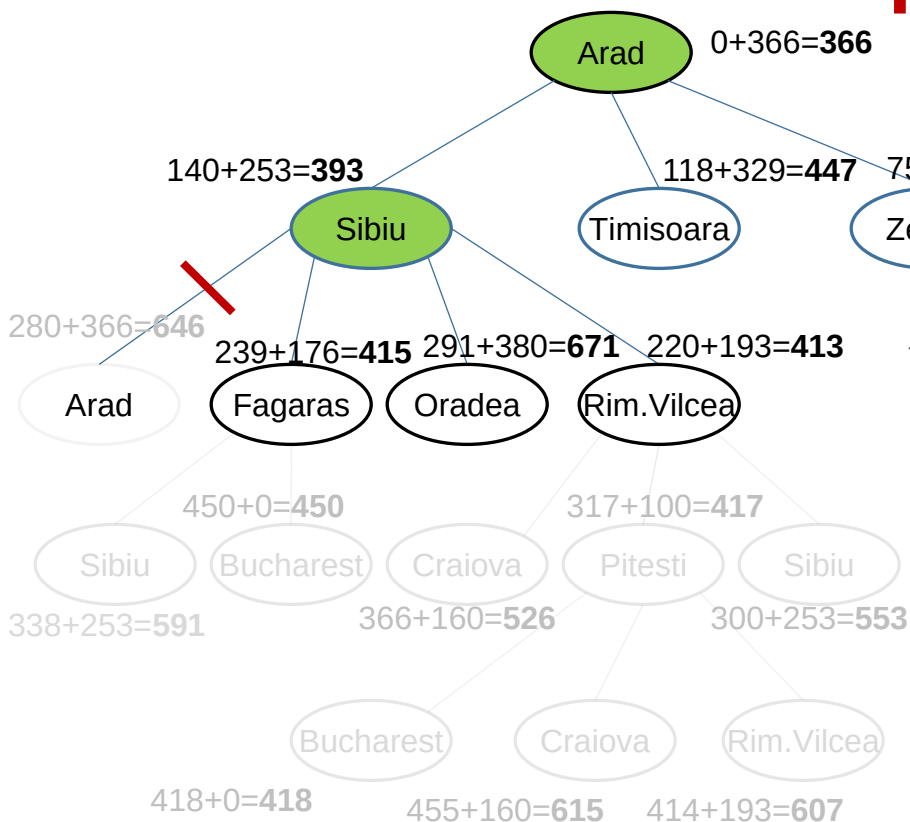


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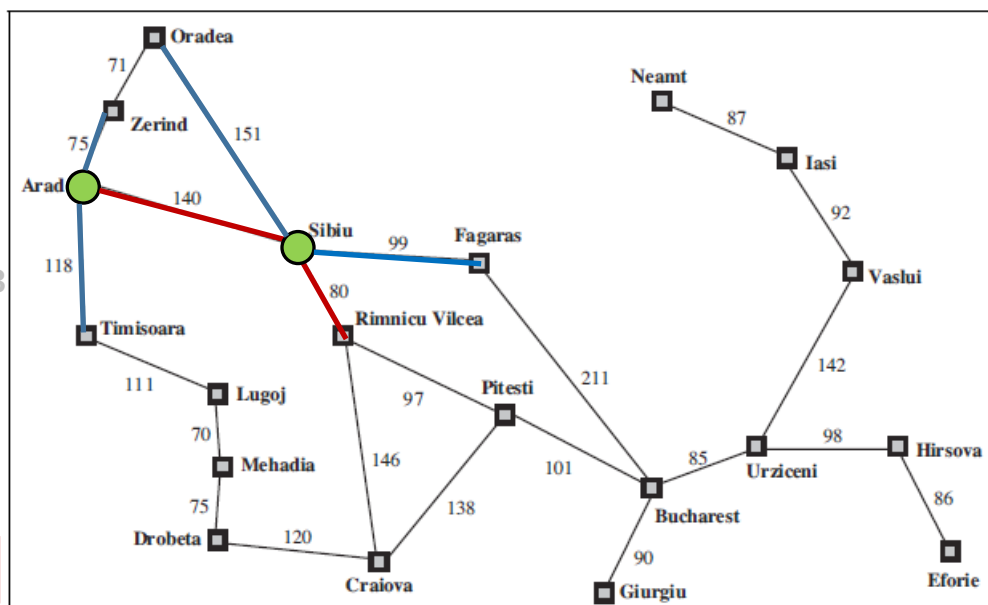
Ex : The Romania Map



EXPLORED SET: Arad, Sibiu

QUEUE

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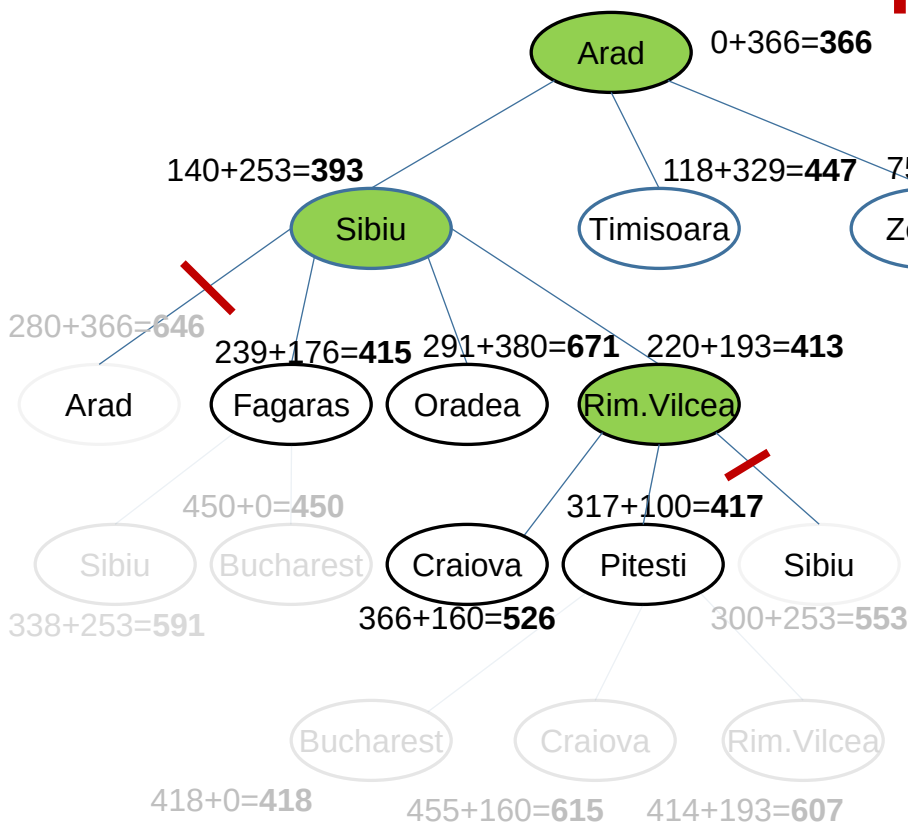


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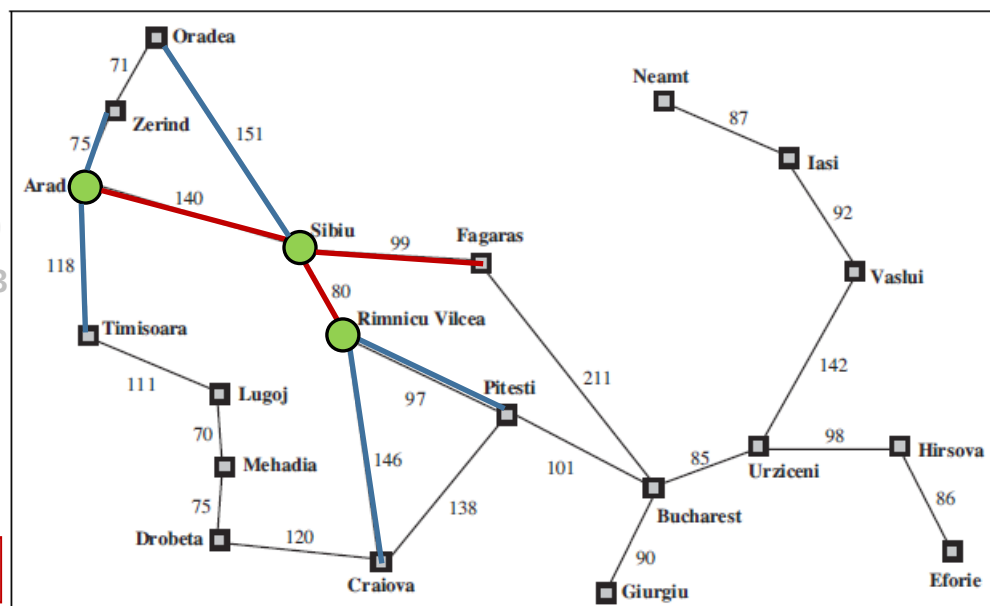
Ex : The Romania Map



EXPLORED SET: Arad, Sibiu, Rim.Vil.

QUEUE

[Arad₃₆₆]
[Sibiu₂₅₃, Timisoara₃₂₉, Zerind₃₇₄]
[Rim.Vilcea₄₁₃, Fagaras₄₁₅, Timisoara₄₄₇, Zerind₄₄₉, Oradea₆₇₁]
[Fagaras₄₁₅, Pitesti₄₁₇, Timisoara₄₄₇, Zerind₄₄₉, Craiova₅₂₆, Oradea₆₇₁]

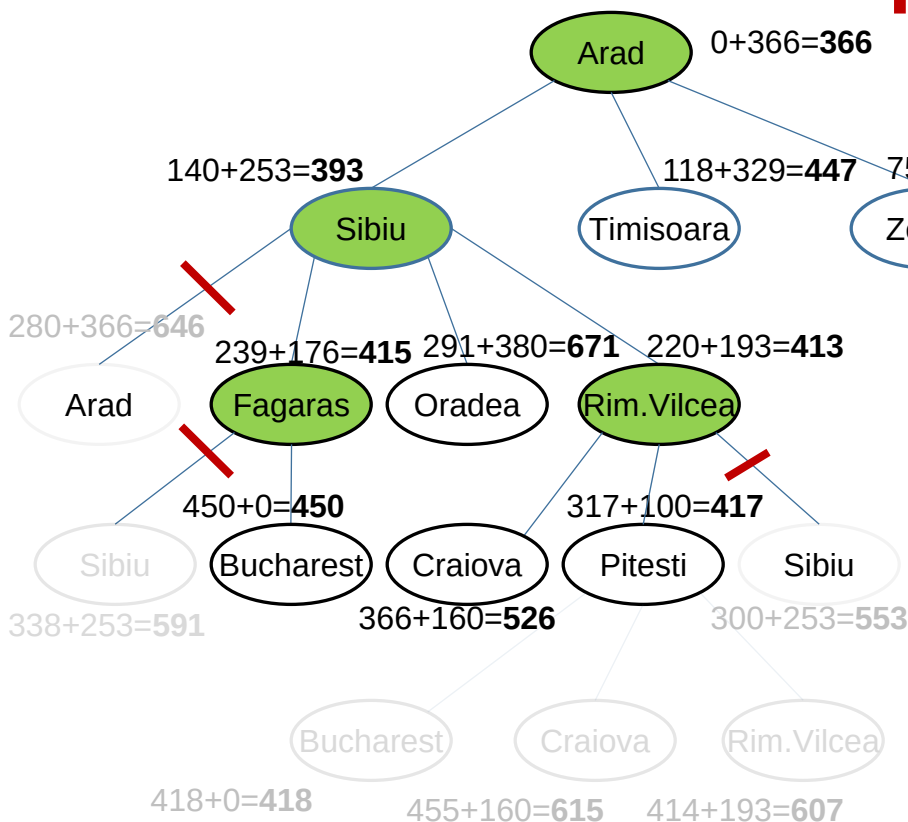


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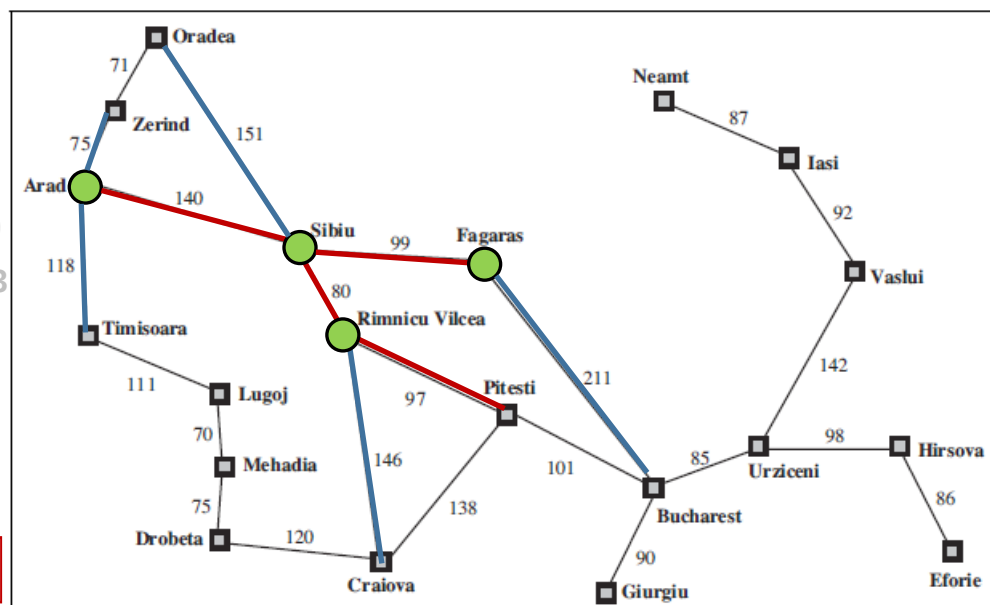
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EXPLORED SET: Arad, Sibiu, Rim.Vil., Fagaras

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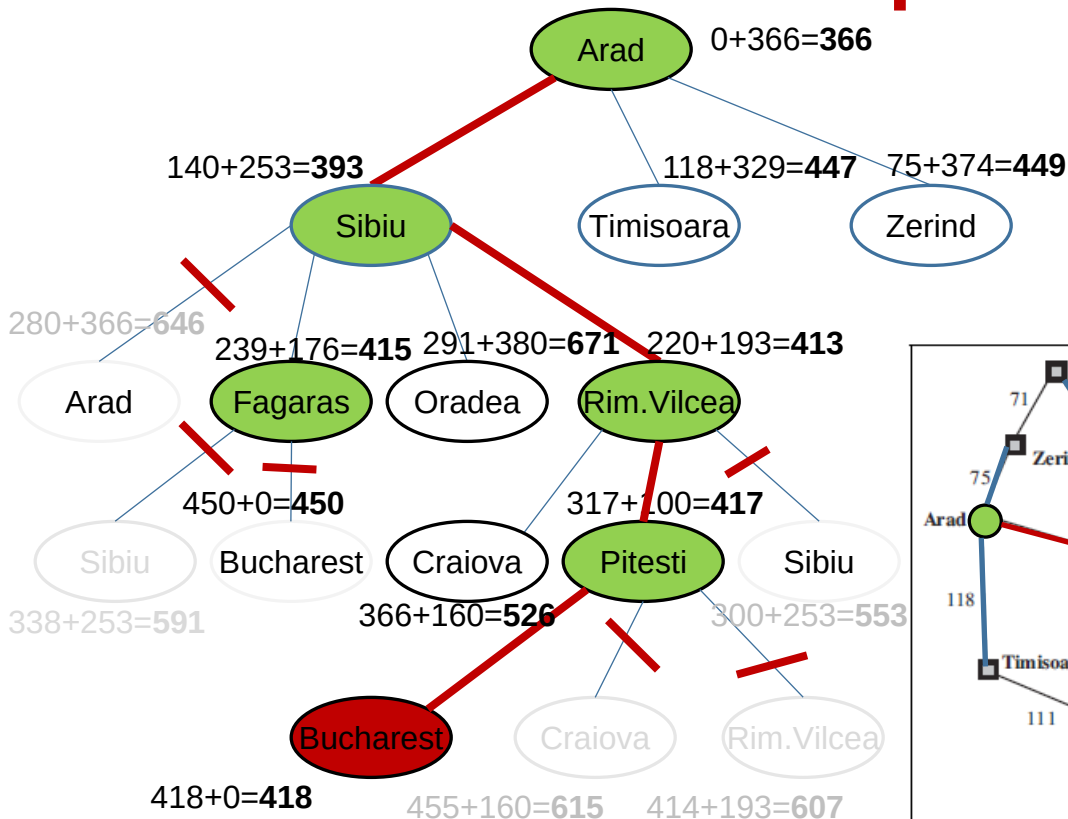


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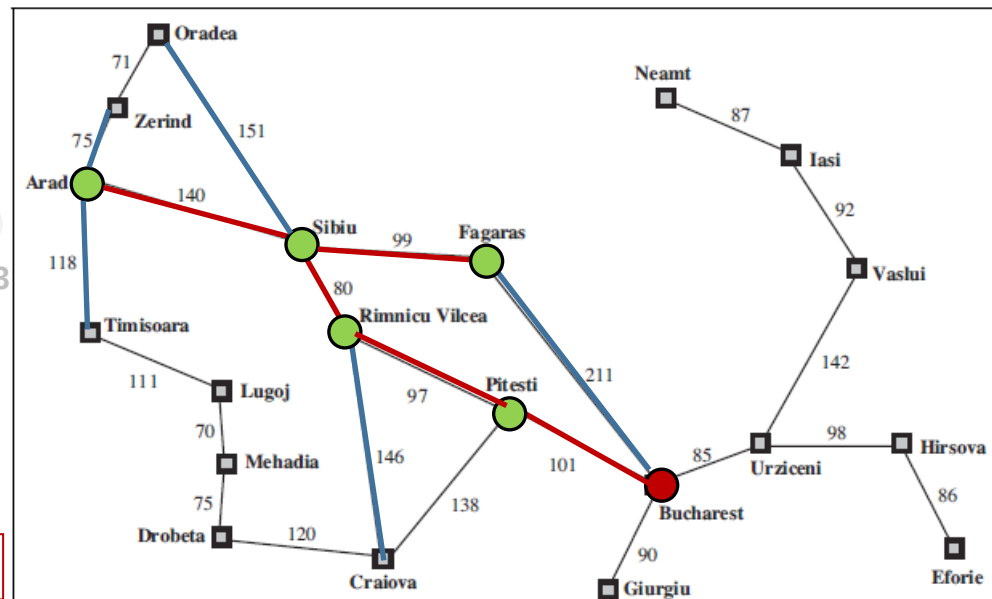
Ex : The Romania Map



EXPLORED SET: Arad, Sibiu, Rim.Vil., Fagaras, Pitesti

QUEUE

[Arad₃₆₆]
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Performance Measure for A* Search

A* search algorithm is **complete** if all step costs exceed some finite $\epsilon > 0$ and if b is finite (like uniform-cost search).

A* search strategy is **optimal** if the heuristic function $h(n)$ satisfies the following conditions:

- **Admissibility**
- **Consistency**

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Conditions for optimality of A* Search:

- Admissibility
- Consistency

An **admissible** heuristic is one that never overestimates the exact cost to reach the goal.

$$h(n^{(k)}) \leq \sum_{t=2}^k c(n^{(t)}, a_t, n^{(t-1)})$$

for any node $n^{(k)}$ where $n^{(1)}$ is the goal node.

$$f(n) = g(n) + h(n)$$

If $h(n)$ is admissible, $f(n)$ never overestimates the true cost of a solution along the current path through n .

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If the following condition (triangle inequality) holds, then $h(n)$ is said to be **consistent**:

$$h(n^{(k)}) \leq c(n^{(k)}, a_k, n^{(k-1)}) + h(n^{(k-1)})$$

for every node $n^{(k)}$ and any action a_k .

Every consistent heuristic is also admissible (**prove it**), but an admissible heuristic rarely may be inconsistent.

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The **tree-search version** of A^* is **optimal** if $h(n)$ is **admissible**, while the **graph-search version** is **optimal** if $h(n)$ is **consistent**.

Since every consistent heuristic is also admissible, it is sufficient to prove optimality of A^* for the consistency.

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The Proof of Optimality

1. If $h(n)$ is consistent, then the values of $f(n)$ along any path are nondecreasing.

Proof: Suppose node n' is a successor of node n for some action a ; then

$$g(n') = g(n) + c(n, a, n')$$

and we have

$$\begin{aligned} f(n') &= g(n') + h(n') \\ &= g(n) + c(n, a, n') + h(n') \quad (h(n) \leq c(n, a, n') + h(n')) \\ &\geq g(n) + h(n) = f(n) \\ &\rightarrow f(n') \geq f(n) \end{aligned}$$

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The Proof of Optimality

2. Whenever A^* selects a node n for expansion, the optimal path to that node has been found.

Proof: $f(n)$ along any path is nondecreasing and the algorithm selects minimum cost valued $f(n)$ of a node n for expansion.

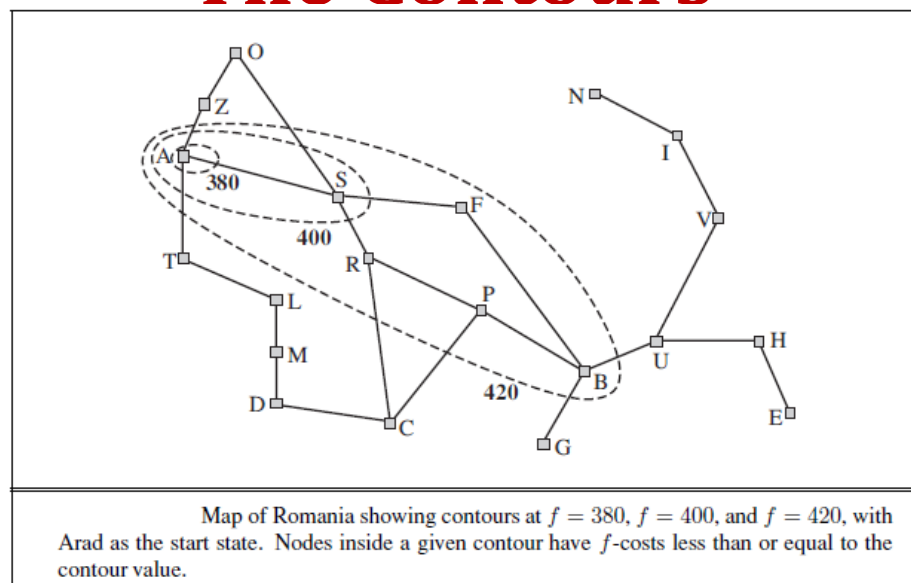
From the two preceding observations, the first goal node selected for expansion must be an optimal solution because f is the true cost for goal nodes (which have $f = g$) and all later goal nodes will be at least as expensive.

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The Contours



The fact that f -costs are nondecreasing along any path also means that we can draw **contours** in the state space. For example, inside the contour labeled 400, all nodes have $f(n)$ less than or equal to 400.

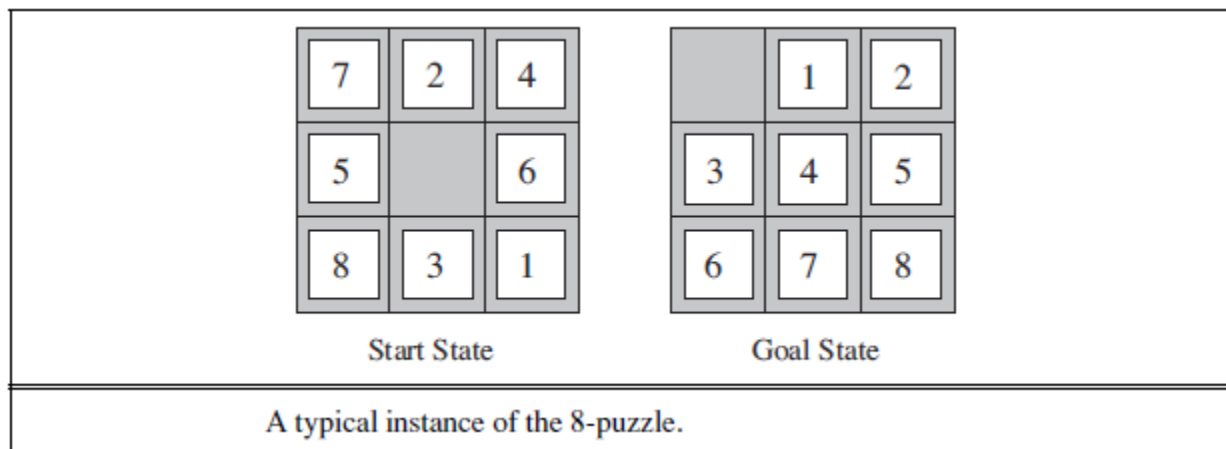
With **uniform-cost search** (A^* search using $h(n) = 0$), the bands will be “**circular**” around the start state. With more accurate heuristics, the bands will **stretch toward the goal state and become more narrowly focused around the optimal path**.

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Heuristic Functions



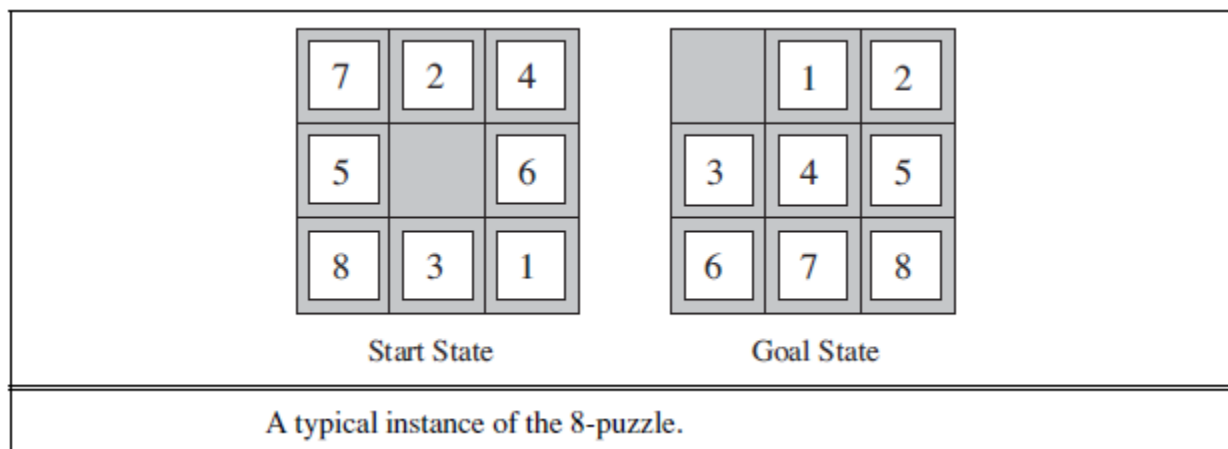
Find appropriate heuristic functions h for the 8-puzzle!!!

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Heuristic Functions



h_1 = the number of misplaced tiles

$$h_1(\text{start state}) = 8$$

$$h_1(\text{goal state}) = 0$$

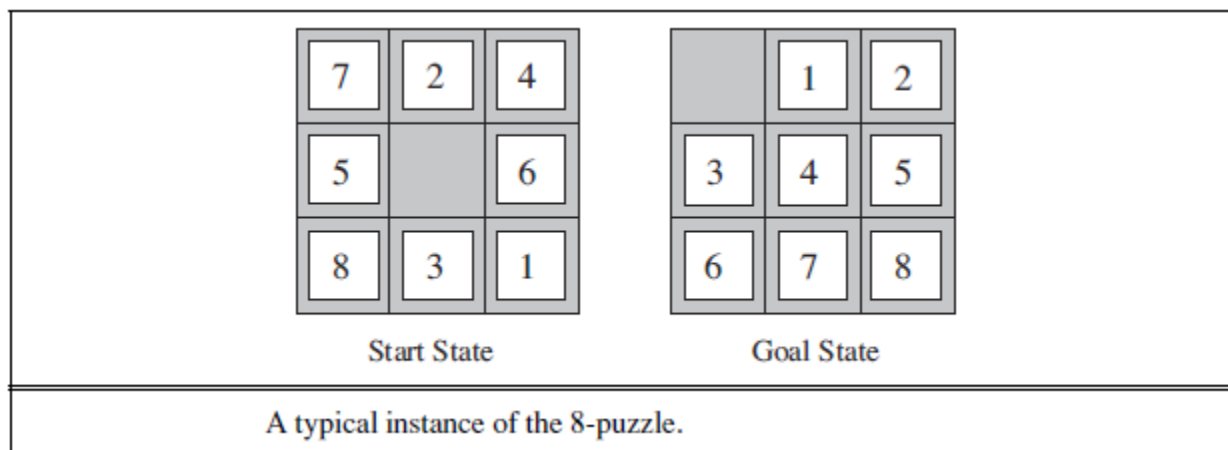
Is h_1 admissible (or consistent) heuristic function? Why?

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Heuristic Functions



h_2 = the sum of the distances of the tiles from their goal positions (Manhattan distance)

$$h_2(\text{start state}) = 3 + 1 + 2 + 2 + 2 + 3 + 3 + 2 = 18$$

$$h_2(\text{goal state}) = 0$$

Is h_2 admissible (or consistent) heuristic function? Why?

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The effect of heuristic on performance

One way to characterize the quality of a heuristic is the **effective branching factor** b^* . If the total number of nodes generated by A^* for a particular problem is N and the solution depth is d , then b^* is the branching factor that a uniform tree of depth d would have to have in order to contain $N+1$ nodes. Thus,

$$N + 1 = 1 + b^* + (b^*)^2 + \dots + (b^*)^d$$

For example, if A^* finds a solution at depth 5 using 52 nodes, then the effective branching factor is 1.92.

A **well designed heuristic** should have a value of b^* close to 1.

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The effect of heuristic on performance

	Search Cost (nodes generated)			Effective Branching Factor		
d	IDS	$A^*(h_1)$	$A^*(h_2)$	IDS	$A^*(h_1)$	$A^*(h_2)$
2	10	6	6	2.45	1.79	1.79
4	112	13	12	2.87	1.48	1.45
6	680	20	18	2.73	1.34	1.30
8	6384	39	25	2.80	1.33	1.24
10	47127	93	39	2.79	1.38	1.22
12	3644035	227	73	2.78	1.42	1.24
14	–	539	113	–	1.44	1.23
16	–	1301	211	–	1.45	1.25
18	–	3056	363	–	1.46	1.26
20	–	7276	676	–	1.47	1.27
22	–	18094	1219	–	1.48	1.28
24	–	39135	1641	–	1.48	1.26

Comparison of the search costs and effective branching factors for the ITERATIVE-DEEPENING-SEARCH and A^* algorithms with h_1 , h_2 . Data are averaged over 100 instances of the 8-puzzle for each of various solution lengths d .

To test the heuristic functions h_1 and h_2 , 1200 random problems with solution lengths from 2 to 24 (100 for each even number) are generated and are solved with **iterative deepening search** and with **A^* search** using both h_1 and h_2 . The figure gives the average number of nodes generated by each strategy and the effective branching factor.

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The effect of heuristic on performance

Is h_2 is always better than h_1 ?

Yes, because...

$$h_2(n) \geq h_1(n) \quad (h_2 \text{ dominates } h_1)$$

Every node n with $f(n) < C^*$ will surely be expanded if C^* is the cost of the optimal solution path ($f(n) = g(n)$ on the goal node). This means that

$$\begin{aligned} f(n) &= g(n) + h(n) < C^* \\ h(n) &< C^* - g(n) \end{aligned}$$

Since $h_2(n) \geq h_1(n)$ for all nodes, every node that is surely expanded by A* search with h_2 will also surely be expanded with h_1 , and h_1 might cause other nodes to be expanded as well.

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The effect of heuristic on performance

It is **generally better** to use a heuristic function **with higher values**, provided it is consistent and that the computation time for the heuristic is not too long.

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Generating admissible (also consistent) heuristics from relaxed problems

Is it possible for a computer to invent heuristics mechanically?
How can we invent a new heuristic?

Let us define **relaxed problem** as a problem with fewer restrictions on the actions. Then, any optimal solution in the original problem is also a solution to the relaxed problem; but the relaxed problem may have better solutions.

Hence, **the cost of an optimal solution to a relaxed problem is an admissible heuristic (it is also consistent) for the original problem.**

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Generating admissible (also consistent) heuristics from relaxed problems

If a problem definition is written down in a formal language, it is possible to construct relaxed problems automatically. For example, if the 8-puzzle actions are described as:

A tile can move from square A to square B if A is horizontally or vertically adjacent to B and B is blank,

We can generate three relaxed problems by removing one or both of the conditions:

- (a) A tile can move from square A to square B if A is adjacent to B.
- (b) A tile can move from square A to square B if B is blank.
- (c) A tile can move from square A to square B.

From (a), we can derive h_2 (Manhattan Distance).

From (c), we can derive h_1 .

Discuss heuristic (known as Gaschnig's heuristic) derived from (b). Compare with other heuristics.

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Generating admissible (also consistent) heuristics from relaxed problems

If a collection of admissible heuristics $h_1, h_2, h_3, \dots, h_m$ is available for a problem and a heuristic h_t , $t \in \{1, 2, \dots, m\}$ dominates any of the others, we can choose h_t . But, if none of them dominates any of the others, then we can define a new heuristic

$$h(n) = \max\{h_1(n), h_2(n), \dots, h_m(n)\}$$

Here for each node, the heuristic that has maximum value is selected. Because the component heuristics are admissible, h is admissible. Furthermore, h dominates all of its component heuristics.

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The generalization of A* Approach

$$f(n) = \alpha g(n) + (1 - \alpha)h(n)$$

Here is α a parameter and $0 \leq \alpha \leq 1$.

$\alpha=0.5$ $\longrightarrow$ A* search

$\alpha=1.0$ $\longrightarrow$ Uniform-cost Search

$\alpha=0.0$ $\longrightarrow$ Greedy Search

The value of α can be changed during the solution.

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Main Reference

Stuart Russell, Peter Norvig; “Artificial Intelligence A Modern Approach”, Pearson, 3rd Edition, 2013.